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To cite this article: Nipada Ruankaew, Christophe Le Page, Pongchai Dumrongrojwattana, Cécile Barnaud, Nantana Gajasen, Annemarie van Paassen & Guy Trébuil (2010) Companion modelling for integrated renewable resource management: a new collaborative approach to create common values for sustainable development, *International Journal of Sustainable Development & World Ecology*, 17:1, 15-23, DOI: [10.1080/13504500903481474](https://doi.org/10.1080/13504500903481474)

To link to this article: <https://doi.org/10.1080/13504500903481474>



Published online: 12 Feb 2010.



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Companion modelling for integrated renewable resource management: a new collaborative approach to create common values for sustainable development

Nipada Ruankaew^{a,b,*}, Christophe Le Page^{b,c}, Pongchai Dumrongrojwattana^{a,d}, Cécile Barnaud^{b,c}, Nantana Gajasen^{a,b}, Annemarie van Paassen^e and Guy Trébuil^{b,c}

^aDepartment of Biology, Faculty of Science, Chulalongkorn University, Bangkok, Thailand; ^bCU-CIRAD ComMod Project, Chulalongkorn University, Bangkok, Thailand; ^cCIRAD, UPR GREEN, Montpellier, F-34398, France; ^dDepartment of Geography, Paris X University, Nanterre, Hauts de Seine, France; ^eDepartment of Communication Science, Communication & Innovation Group, Wageningen University, Wageningen, The Netherlands

The sustainable management of renewable resources is often complicated by the diversity and dynamic nature of the ecological and socio-economic systems involved. As the dynamics and interactions of these systems are highly complex and frequently unpredictable, there is a need to opt for transdisciplinary research addressing adaptive and integrated management of renewable resources. Companion modelling (ComMod) is a multi-agent systems (MAS)-based approach relying on synergistic effects between role-playing games (RPG) and agent-based models (ABM) to facilitate collective information sharing and learning, and to improve coordination among stakeholders for negotiation and decision-making. The iterative and adaptive sequences of fieldwork with modelling activities allows the mutual and interactive participation of stakeholders during the design, implementation, calibration and validation steps of the models, as well as exploration of possible scenarios. ComMod was implemented in a study of the conflict between two ethnic communities and a newly proposed national park in Northern Thailand. Deforestation, biodiversity conservation and the livelihoods of the villagers were key issues discussed during RPG sessions with small groups of participants, and subsequently represented in an ABM simulator used with a larger audience. Consequently, the local stakeholders learned about agro-ecological and socio-economic dynamics and gained an increased awareness of these key issues. Mutual understanding was improved, and the importance of collaborative discussion, essential to negotiations and decision-making, became obvious. Finally, this Northern Thailand experience has shown that collaborative interactions mediated by ComMod were supportive of improved communication and joint learning for the adaptive and integrated sustainable management of renewable resources.

Keywords: companion modelling; Northern Thailand; participatory approach; renewable resources; sustainable management

Introduction

The sustainable management of renewable resources inevitably involves not only ecological dimensions but also the social, economic, cultural and political aspects of the use of resources. Successful management of renewable resources is, therefore, often complicated by the diversity of the interconnected ecological and socio-economic systems and their interactions, as well as by the increasing number of stakeholders concerned with the collective management of common-pool resources and environmental problems. In addition, the dynamic nature of interactions among diverse factors at various levels and scales frequently leads to highly complex, non-linear and divergent processes and the emergence of new phenomena, which are often unpredictable (Liu et al. 2007; Trébuil 2008; van Paassen et al. 2008). As the rate of changes accelerates and the uncertainty increases, there is a need to opt for transdisciplinary research addressing truly adaptive and integrated management of renewable resources (Berkes and Folke 1998; Holling 2001).

Companion modelling (ComMod) is a multi-agent systems (MAS)-based approach that relies on the synergistic effects between role-playing games (RPG) and agent-based models (ABM) to facilitate information sharing, collective learning and exchange of perceptions on a given concrete

issue among researchers and other stakeholders. ComMod can be used either to foster a better understanding of a complex issue or to support negotiation and collective decision-making among stakeholders facing a common renewable resource management problem (Bousquet et al. 1999). The iterative and adaptive sequences of fieldwork and modelling activities allows the mutual and interactive participation of stakeholders during the design, implementation, calibration and validation steps of the models, as well as their joint use to explore possible future scenarios of interest identified by the model end users.

This article presents the theoretical and experimental aspects of ComMod, followed by a case study that draws on a ComMod experiment conducted in Northern Thailand. The results are then discussed, and the potential for collaborative research and sustainable development of renewable resources is explored.

Methods

Companion modelling (ComMod)

ComMod is a MAS-based approach, in which a collection of heterogeneous, autonomous entities, or agents, interacts with each other and with their common environment

*Corresponding author. Email: nipada.r@chula.ac.th

(Gilbert 2008). MAS models are thus adapted to describe dynamic and complex systems of renewable resource management and are able to capture emergent phenomena arising from interactions amongst the diverse dimensions of resources and stakeholders in the systems (Janssen 2003). ComMod is accomplished by the combined use of RPG and ABM, both of which are MAS-based tools (Bousquet et al. 2002; Barreteau 2003).

ComMod has two major aims, (i) to facilitate collective information sharing and learning and (ii) to improve the coordination among stakeholders for negotiation and decision-making (Bousquet et al. 1999; Barreteau 2003). It has been applied to explore various renewable resource management issues in many places around the world, including some 15 sites in South and Southeast Asia¹ (Bousquet et al. 2005). As a relatively new collaborative research and modelling approach, ComMod can be used in two contexts; first to understand the functioning of a complex socio-ecosystem by stimulating dialogue and collective learning, and second to facilitate negotiation and decision-making, and thereby strengthening the capacity of local stakeholders for adaptive resource management. The two contexts are complementary and are not mutually exclusive, as the first objective needs to be fulfilled before proceeding to the second. Most ComMod exercises have achieved the first objective of gaining greater knowledge (Mathevet et al. 2003; Barnaud et al. 2008a; Dumrongrojwatthana et al. 2008; Dung et al. 2008; Naivinit et al. 2008). However, only a few case studies have reached the second objective, and so contributed towards adaptive and integrated management of renewable resources (D'Aquino et al. 2003; Gurung et al. 2006; Barnaud et al. 2008b; Promburom and Bousquet 2008).

The modelling process of ComMod is divided into four successive phases: (i) definition and characterization of the problem, including stakeholder and institutional analysis, based on the existing knowledge and specific surveys to fill the gaps; (ii) conversion of this pool of knowledge so obtained to formal models through participation of the stakeholders in the co-construction of such conceptual models; (iii) implementation of concrete models that can be run (simulators); and (iv) participatory simulations to explore various scenarios of possible solutions identified by the stakeholders (Bousquet et al. 2005). The iterative and adaptive sequences of fieldwork and modelling activities allow inclusive and interactive participation of the stakeholders throughout successive cycles of the ComMod process that are driven by the stakeholders' needs. The approach considers that the points of view of all local stakeholders on the issue to be examined are legitimate. Therefore, the collaborative modelling process welcomes the involvement of all concerned parties in problem definition, model design, implementation, calibration and validation steps of RPG and/or ABM models (Dray et al. 2006). The exchange of perceptions on various aspects of the common issue among local stakeholders and researchers is facilitated by the use of the RPG and/or ABM simulators with the aim of achieving an agreed-upon and shared representation of the problem to be mitigated collectively.

Based on initial conceptual models, RPG are MAS models implemented into playable games, usually with the help of boards, maps, cards and papers, and are often computer-assisted. During RPG sessions, participants enact the actual use and management of renewable resources by playing various roles (frequently their own roles) of resource users or managers. Charts, graphs and tables of relevant ecological and socio-economic indicators are provided for observation of the game results. This experience allows the stakeholders to interact and examine complex systems that they are part of but, importantly, in a non-threatening and friendly atmosphere; this usually triggers discussions amongst them about their common problems and possible solutions. RPG are consequently efficient in gathering information on selected topics (Bousquet et al. 2007). To a certain extent, as the participants share their existing common knowledge of the actual features of the problem at stake, the models can be rather simple and focus only upon critical interactions, for instance, improvement of management through negotiation and new rules (Bousquet et al. 2002; Barreteau et al. 2003).

Complementary to RPG, ABM generate computer simulations that integrate information and indigenous knowledge gathered from RPG sessions, and consequently provide common representations of the complex dynamics of the systems. As they are based on the same MAS structure, the agents and the environment in the ABM are clearly related to the RPG players and game boards, providing parallel representations of the actual resources and users along with the rules and the context of the real system. Whereas all decisions in RPG are human-made, the decisions in ABM are made by the computer executing algorithms specified as instructions written in programming language. Gradually built during the whole ComMod processes, the ABM are mainly used to simulate relevant scenarios selected by the stakeholders to help them explore alternatives in resource management. In particular, they are able to see and discuss their possible specific effects on the resources, normally through observation of relevant ecological indicators. Additionally, stakeholders are also able to examine the specific effects of simulated scenarios on the concerned parties, for example, with social and/or economic indicators showing the benefits and disadvantages (Etienne et al. 2003; Barnaud et al. 2008a).

Playing RPG also helps local stakeholders understand the structure and operation of the computer-based ABM and gives them an opportunity to validate, criticize and improve it (Barnaud et al. 2006). While the RPG tools are usually popular with local stakeholders, they are normally limited to a playable size of 10–20 participants. The game sessions tend to become more complex and cumbersome as the number and diversity of participating stakeholders increases. In addition, RPG sessions are rather costly to organize in terms of the time (half a day per gaming session with 10–20 players), as well as the financial and human resources needed to prepare, moderate, observe and analyse the gaming sessions. Computer-based simulations with an ABM tracing the rules of the RPG are more manageable and time-efficient

(Barreteau et al. 2001; Bousquet et al. 2002; Barreteau 2003). The ABM tools facilitate the transfer of the RPG experience to larger audience groups such as the whole village population (100–200 people) and to decision- or policy-makers at a higher institutional or social hierarchy level who normally have limited time. Accordingly, ComMod researchers usually associate RPG with ABM simulators able to ‘play the game’ in a time- and cost-efficient way, leaving plenty of time for the stakeholders to debate the simulation results and think about new scenarios to be assessed.

More often than not, RPG sessions followed by participatory ABM simulations lead to improved knowledge and understanding, as well as increased dialogue through collective interactions and joint learning. In addition, new aspects, key questions or entirely different problems frequently emerge that can be examined through similar modelling and simulation activities during further field workshops in a new ComMod cycle. Thus the iterative nature of ComMod provides an adaptive approach for the exploration of rapidly evolving renewable resource management issues (Barnaud et al. 2006). For example, ComMod researchers have successfully used collaborative ABM simulations with illiterate Akha ethnic highlanders to explore the rules of credit allocation among different types of households in perennial cash cropping in Northern Thailand (Barnaud et al. 2008a).

Case study on the management of an upper watershed in Northern Thailand

A case study of the ComMod approach dealing with integrated renewable resource management in Northern Thailand is presented here. This case study combined RPG and ABM simulation experiments to raise stakeholders’ awareness and understanding of the effects of the establishment of a recently proposed national park on local farming activities in the upper watershed of Nan Province, Northern Thailand (Barnaud et al. 2008b). Ban Nam Ki and Ban Nam Paeng villages are two ethnic Mien (or Yao) communities located along the border of the new Nanthaburi National Park (NNP) in Western Tha Wang Pha District, Nan Province. Their livelihoods until the 1970s depended upon shifting cultivation-type agriculture of maize, upland rice and opium poppies. In the late 1970s, significant deforestation occurred due to commercial logging concessions and, to a certain extent, extensive cash crop cultivation being forced on the villagers by the Thai government to replace opium poppy cultivation. To combat the serious decline of forest cover in this head-water area, the Royal Forest Department (RFD) delimited agricultural land, thereby reducing available farmland to the villagers. The Mien villagers had to replace their shifting cultivation practices with the permanent cultivation of annual (mainly maize for animal feed) and perennial (mainly lychee orchards) commercial crops. Therefore, subsistence livelihoods were replaced with new and far more unequal socio-economic situations, in which the villagers depended upon outside markets, both for inputs and outputs.

As a result of these transformations, the most resource-poor villagers were seriously in debt, and had to supplement

their income with off-farm employment in urban areas, and collection of non-timber forest products (NTFP), such as bamboo shoots and high-value Arenga palm fruits. In 1996, the establishment of the NNP was proposed in the area, although it had not yet been officially declared at the time of the study (2006). Park boundaries were not clearly defined, leading to the Mien villagers’ confusion and frustration over the possibility of losing already limited farmland. Moreover, specific forest management rules and regulations were not known, especially regarding the rights of the villagers to continue gathering NTFP, which are important sources of income for the most economically vulnerable households. In the two study villages, a conflict of interest between the villagers and the national park was evident. A lack of communication between the concerned stakeholders did not help in mitigating the conflict, and led to widespread mistrust and misunderstanding among the stakeholders. The conflict over the utilization of forest resources in the study area encompassed both ecological and social dimensions, resulting in intricate situations. Forest cover of the upper watershed, water availability and biodiversity conservation were issues as important as dependence of the villagers’ livelihoods upon farming and NTFP.

In 2006, the ComMod approach was applied to improve the communication among the conflicting parties. The challenge was to facilitate communication between the Mien villagers and the national park administrators in order to improve their understanding of the complex issue of forest resource management and to lead to a new set of rules regulating the exploitation of NTFP by the economically- and resource-poor villagers. The ComMod exercise combined modelling activities in the laboratory, namely the design of two successive RPG and of an ABM simulator, with a series of field workshops, including RPG gaming sessions and subsequent interviews, in the two villages (Figure 1). The first set of workshops was followed by a presentation of the ABM simulator and the stakeholders’ feedback from field workshops to the NNP officers. A final joint workshop was implemented with the participation of the farmers from both villages as well as the RFD and NNP officers.

During the initial diagnosis, based on secondary data analysis, landscape analysis and individual interviews with local farmers in early 2006, the research team was able to identify a key renewable resource management problem, i.e. the possible conflict between the NNP and the two villages over forest utilization in the midst of the newly proposed national park demarcation, and the main stakeholders involved. The initial diagnosis also revealed a socio-economic diversity among the local farming households, and distinguished three types of household based on the relative importance of farm-based and off-farm income sources, as well as their dependence on NTFP.

The first type of local household represented poor families with insufficient income due to small land parcels that normally cultivated low-value annual crops (maize and upland rice). They were thus highly dependent on income

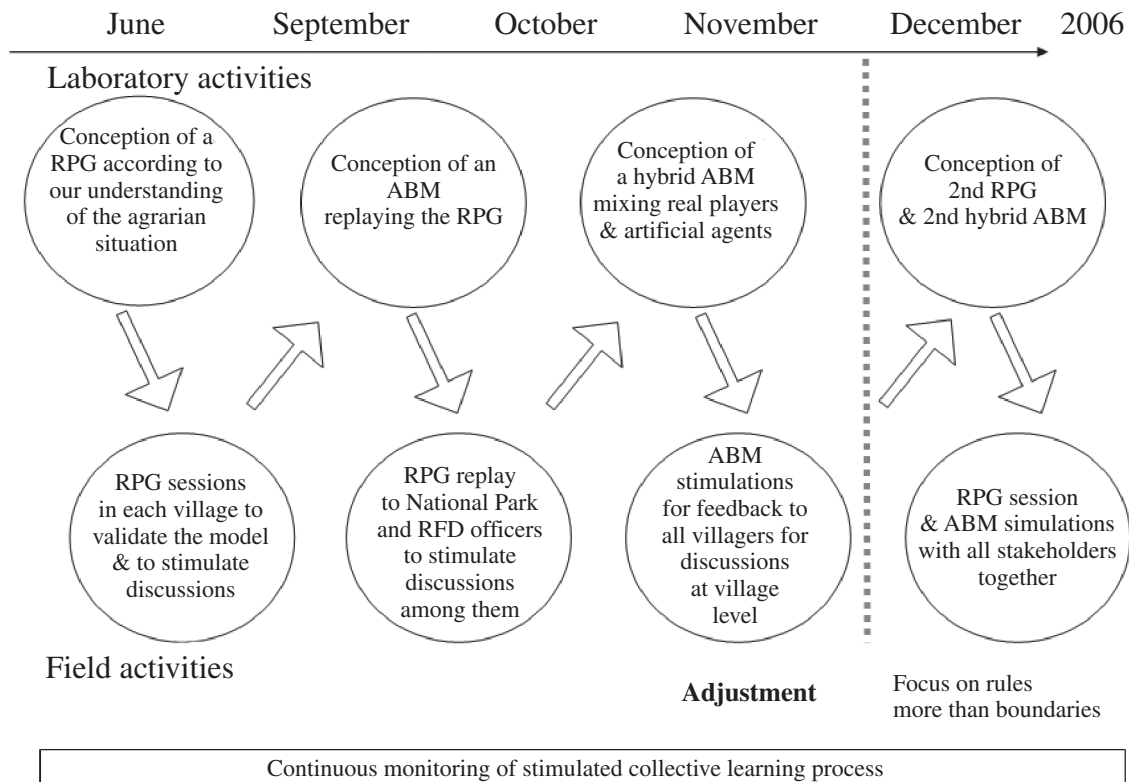


Figure 1. Iterative companion modelling process alternating field and laboratory activities implemented in Nan Province, Northern Thailand (RPG, role-playing game; ABM, agent-based model).

from selling Arenga fruits, consumption of other NTFP and low-wage off-farm employment.

The second type represented medium-income households owning larger land parcels with limited capital to invest in growing a high-value perennial crop (lychee) together with annual crops. NTFP offered them an important additional source of income, especially when their crops produced a low harvest or income for other reasons.

Finally, the third type of local household included well-off families with larger land areas and relatively more capital to invest in high-value lychees, although the majority of their income came from profitable off-farm income, such as selling soymilk in bigger towns (Barnaud et al. 2008b).

This preliminary diagnostic analysis showed how crucial NTFP, in particular Arenga fruits, were to the economic survival of the most resource-poor farming households. Accordingly, the ComMod activities were designed to focus on the effects of farming and forest utilization on the livelihoods of villagers belonging to different socio-economic circumstances.

The first round of participatory workshops, which included RPG sessions, subsequent group discussions and individual interviews, was held in June 2006 with selected farmers from the two villages representing the socio-economic diversity of the households. In each village, 12 farmers played with a board representing farm areas and forests (Figure 2a). The players needed to make decisions to allocate labour force and money, choose crop types (annual and/or perennial) and to collect Arenga palm fruits and other

NTFP from the forest. After each round of play, the players were presented with their crop yields and incomes gained. Subsequent discussions led to changes in the game rules and possible scenarios to explore, in this case, with and without the presence of a national park. In the scenario with the national park, a rule was stipulated that the villagers could neither farm nor collect NTFP in the land beyond the park boundary (Barnaud et al. 2008b).

The key issues and important forest–farmland interactions were later represented in a simple ABM simulator exhibiting a virtual space similar to the RPG game board (Figure 2b). Figure 3 shows the unified modelling language (UML) class diagram, with the main entities of the model as rectangular boxes. ‘Farmer’ and ‘NationalPark’ are the agents of the system (hence the term MAS). Both of them interact through the space, in that they are related to the ‘LandUnit’ entity that represents a small homogeneous portion of space characterized by cover type. The ‘Farmer’ agent is characterized by a set of attributes (top part of the box) and a set of operations (lower part of the box). The ‘Forest’ cover type is characterized by the presence of two types of product, ‘Arenga’ and ‘OtherForestProduct’ (e.g., bamboo shoots). The crops are either annual or perennial, each type being characterized by specific values of input cost, gross product and market price. In the model, the ‘NationalPark’ agent does not directly change the land cover; its only action is to define rules for farmers to access the forest portion of the landscape. In the RPG version of this MAS, the players

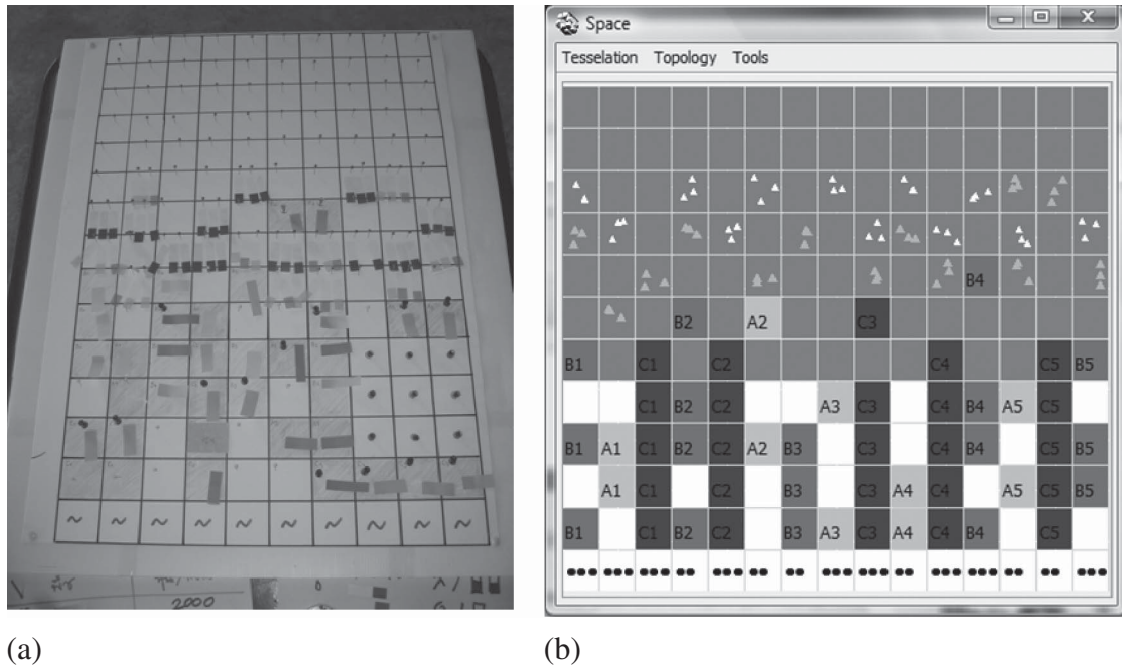


Figure 2. Similarity of features between the main gaming board (a) used in the role-playing game and the interface of the ABM simulator (b) used in the ComMod study of a forest utilization conflict in Nan Province, Northern Thailand.

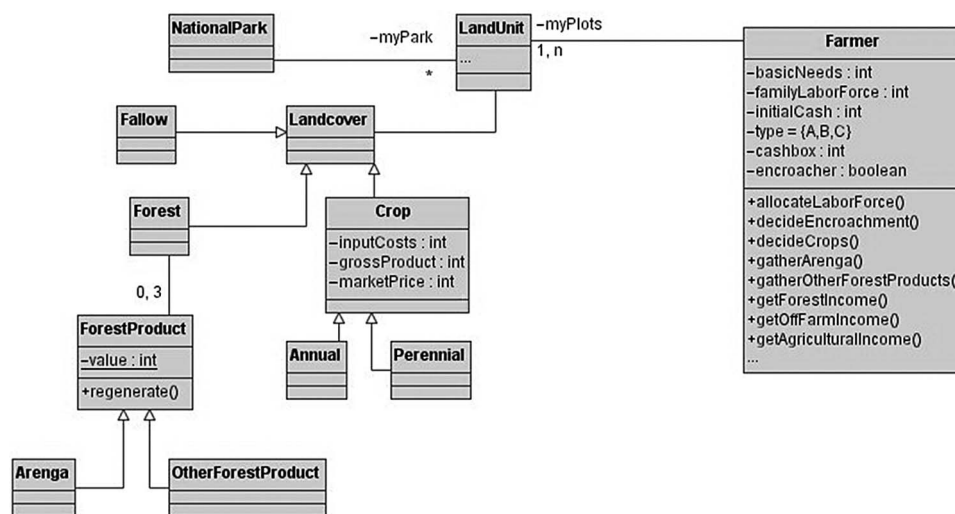


Figure 3. Class diagram showing the structure of the ABM simulator used in the ComMod study of a forest utilization conflict in Nan Province, Northern Thailand.

representing the national park are directly communicating these rules with the players representing the farmers. In the computerized ABM version, this is achieved through the definition of scenarios.

The duration of each simulation scenario was set to 10 time steps (or crop years). The sequence of activities executed at each simulation time step unfolds as follows: (i) the market price for lychees is set either low or high; (ii) each household decides on the allocation of its labour force, depending on cash availability; (iii) each household decides on the type of crop (annual or perennial) in each plot it owns (for the scenario enabling forest encroachment, the number of plots may increase over the 10 time

steps); (iv) depending on cash availability, each household decides whether or not to collect NTFP in the forest; (v) the NTFP regenerate according to a simple rule (Figure 4).

A hybrid version of the ABM simulator was also constructed, with an interface providing the possibility to allow any 'Farmer' agent of the computer model to become an avatar, fully controlled by a participant. For instance, when the hybrid version of the model was used in Ban Nam Paeng, three human players, each playing a different farmer type, played with seven virtual figures, comprising a total of 10 'Farmer' agents. This simulator was able to replay the gaming sessions *in silico* and was used to provide feedback from the gaming sessions played

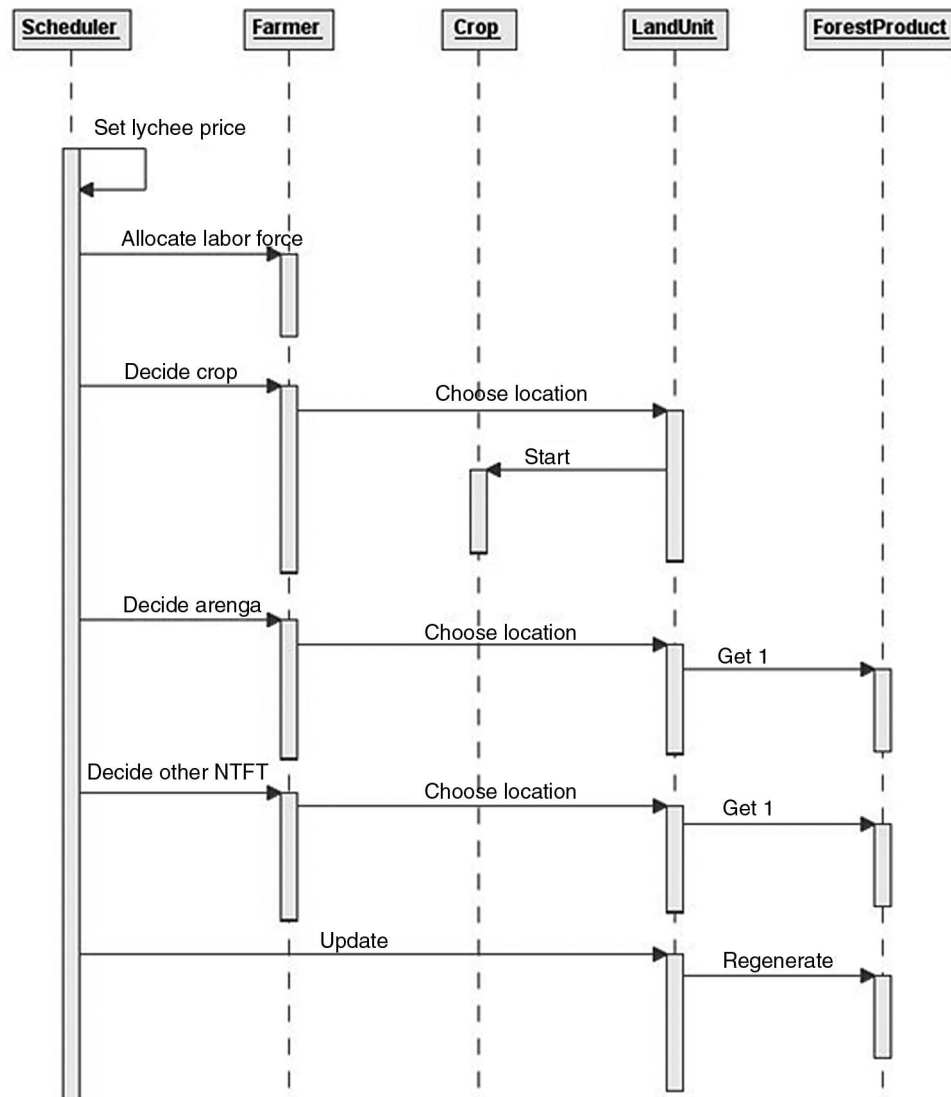


Figure 4. Sequence of activities executed at each simulation time step of the ABM simulator.

by a dozen farmers to the whole village population. It was also used to communicate the villagers' point of view to the RFD and NNP officers before the second round of RPG sessions, held in December 2006, in which the villagers from both villages, the RFD and NPP officers participated.

As part of the ComMod approach, group discussions and individual interviews were routinely conducted with the participants immediately after each of the RPG or ABM simulation sessions, both to record what they learned and received from the ComMod experience, and to gather their suggestions on how to improve the on-going modelling process and its tools. The questions included an assessment of the relationship between actual farming circumstances and their representations in the tools used, levels of understanding of the features and operation of the RPG or ABM tools, and elicitation of the behaviour of the stakeholders towards the issues of resource management, which was the focus of the ComMod sessions.

In June 2007, an external evaluation of this ComMod case study was conducted based on a framework to document and assess the experiences of the process designers and local participants.² The external evaluation, conducted through interviews, aimed to assess the actual implementation and diverse effects of this ComMod process on its participants, both immediate and long term (van Paassen and Patamadit 2008). Combining the routine evaluation and monitoring steps inherent to the ComMod approach with the additional evaluation by external experts has provided an innovative framework for a systematic assessment of this case study.

Results and discussion

In both villages, the players became rapidly at ease with the features and rules of the RPG. Most players chose cropping and off-farm activities as they would in reality according to their socio-economic status. During the scenario with the

presence of the national park, all of the poor and most of the medium-income households could not make sufficient income for basic subsistence needs, due to the sharp reduction of available NTFP. On the other hand, the well-off families were hardly affected financially.

Subsequent plenary discussions and individual interviews showed that different types of local household had unequal access to information and expressed different perceptions regarding the issue of the national park establishment. The poor households, whose livelihoods depended highly on NTFP and whose participation in local politics was limited, were not aware of the upcoming formalization of the national park until the RPG sessions, and expressed their fear of losing access to both farmland and the right to collect NTFP. The medium-income households were more informed about the national park establishment and showed concerns over the risks of losing some farmland, but appeared not to worry about losing NTFP rights. In contrast, the well-off families were generally well informed and expressed no personal interest in the issues. Some of them, however, realized the possible difficulty that other villagers might face (Barnaud et al. 2008b). In this case, the initial diagnosis of the socio-political context enabled the ComMod process to address and understand the heterogeneity and the relationships among villagers of different socio-economic and political conditions, which in turn affected how they made decisions and responded to resource management issues.

The outcome of the ComMod process implemented in the remote villages with people having received limited or no formal education showed that villagers and researchers alike gained a better understanding of the forest resources and the farmers' livelihood issues. A total of 85% of participants said that the RPG increased their awareness of the national park issue (Barnaud et al. 2008b). Furthermore, access rules to NTFP utilization emerged as a more significant problem than the actual national park boundary itself.

The results from the first RPG sessions led to the construction of a hybrid ABM simulator, which represented the same resources, stakeholders and rules of resource utilization. After presentation of the ABM simulator, the NNP officers expressed a better understanding of the objectives and process of ComMod and agreed to participate in the second participatory workshop with farmers from the two villages. As a result, it was agreed that a mutual understanding through better communication, information-sharing and exchange of points of view among different local stakeholders was needed, and a common platform was necessary for mediation and negotiation of the forest resource management issues in the area. This achievement was seen as a first step towards more adaptive and sustainable management of forest resources in this area. However, at the last minute, the chief of the NNP did not join the workshop. In addition, due to the highly top-down regulated nature of the national park law, only informal local agreements could be reached and the villagers remained vulnerable to the looming change in forest utilization rules.

Another problem was that a few households of forest encroachers from the Ban Nam Ki village, being afraid of

having their behaviour publicly exposed, refused to participate in any of the ComMod activities. This underlines the need for a more thorough socio-political diagnosis and sensitizing activities, such as within-village discussions (van Paassen and Patamadit 2008), prior to the field workshops to try to limit the somewhat self-exclusion of key players, as their important participation is required so that the recommendations made at the end of the ComMod process represent a consensus among all stakeholders and have a chance to be implemented and enforced at the village level.

The external evaluation revealed that the ComMod approach was useful to the local villagers for individual learning about the consequences of proposed regulation by the new national park. Specifically, the villagers learned about agro-ecological dynamics and socio-economic equity issues. Several of them increased and used their social networks to communicate about the new national park issues, and realized the essential importance of mutual discussion with other villagers in order to obtain a satisfactory level of negotiation and decision-making processes that involve powerful government agencies. The researchers gained a better understanding of the agro-ecological situation and of the interactions among the local stakeholders. They also found the need to fill a knowledge gap on the ecology of Arenga palms to better represent the production and harvest of this key NTFP. Likewise, the NNP officers gained an increased awareness of the villagers' circumstances and points of view, and acknowledged the need for better communication and coordination mechanisms with the villagers. Most importantly, the ComMod process increased mutual understanding among and between the villagers and with the NNP officers (van Paassen and Patamadit 2008).

In general, the ComMod approach proved to be a strong and practical approach in the representation and demonstration of the stakeholders' behaviours and interests, and in dynamics between the stakeholders and the renewable resources. Although limited to a small number of participants, RPG proved to be an effective tool to reveal individual and collective behaviours that are not easily identified with less interactive tools, such as individual interviews, especially when social tension among stakeholders exists (Barnaud et al. 2008b). However, some stakeholders expressed concerns that the complexity and diversity of issues during the sessions could overwhelm the participants. This suggests a potential limitation of some concern when more diverse types of conflict and stakeholder are included in a single session and, therefore, the need to design and implement ComMod by focusing on a well-defined practical resource management problem faced by the potential participants. Sessions need to be structured in order to obtain focused discussions, and unravel issues one at a time in accordance with their priority (van Paassen et al. 2008).

ComMod was also shown to be useful in multi-level communication. In the present case study, the iterative sessions of RPG sessions and ABM simulator acquainted the local farmers and national park officers, who are used to top-down administration, with the possibility of bottom-up decision-making and scenario-enhanced negotiation.

Specifically, the RPG was successfully used with small groups of local stakeholders, and the accompanying ABM simulator was subsequently used to provide feedback to the whole village community as well as familiarize the national park officers before their participation in the second RPG sessions. This synergistic interaction between the RPG sessions and ABM simulations has consequently provided an original achievement in up-scaling ComMod to include a diverse and hierarchical structure of stakeholders involved in the collective management of renewable resources.

Conclusion

The overall outcome of this Northern Thailand experience has shown that collaborative interactions between researchers and local stakeholders, mediated by ComMod tools, were supportive of enhanced communication, dialogue and joint learning to create common values among the various stakeholders and to improve adaptive and integrated management, which can lead to the sustainable development of renewable resources. However, challenges and obstacles in up-scaling the ComMod processes still need to be explored. The construction and use of an ABM simulator replaying RPG sessions in the case study described here can be seen as a promising step in that direction. This case study also underlines the need for a supportive policy environment to facilitate the implementation of collective action plans emerging from such collaborative modelling processes for the sustainable and integrated management of renewable resources.

Notes

1. Descriptions of ComMod projects in South and Southeast Asia can be found at <http://www.ecole-commmod.sc.chula.ac.th> and <http://www.cpwf25.sc.chula.ac.th>.
2. The external evaluation was part of a large-scale evaluation of some 30 ComMod case studies in various continents, carried out under the Agriculture and Sustainable Development (ADD) Programme of the National Research Agency (ANR), France.

Acknowledgements

The authors thank Chulalongkorn University (CU) for creating an inspiring research environment. They also thank the Challenge Program Water and Food (CPWF) of CGIAR for financial support under Project PN25 for our participation in the second International Sustainability Conference in 2008.

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